Teaching a model to tell roses, sunflowers and tulips apart

Transfer learning with 100 photos per flower

neuefische · AMLE project

A wrong photo match sends a customer the wrong care guide

The situation

> Customers upload a photo of their flower and get a care guide
>
> Staff choose the right guide **by hand** today
>
> Goal: pick it **automatically** for roses, sunflowers and tulips

The question

> Given a photo, **which of the three flowers** is it?
>
> **When can we trust** the answer, and when should staff check it?

We have only 300 photos to learn from, so the model must reuse prior knowledge

300

Training photos
100 per flower

60

Validation photos
to choose the model

30

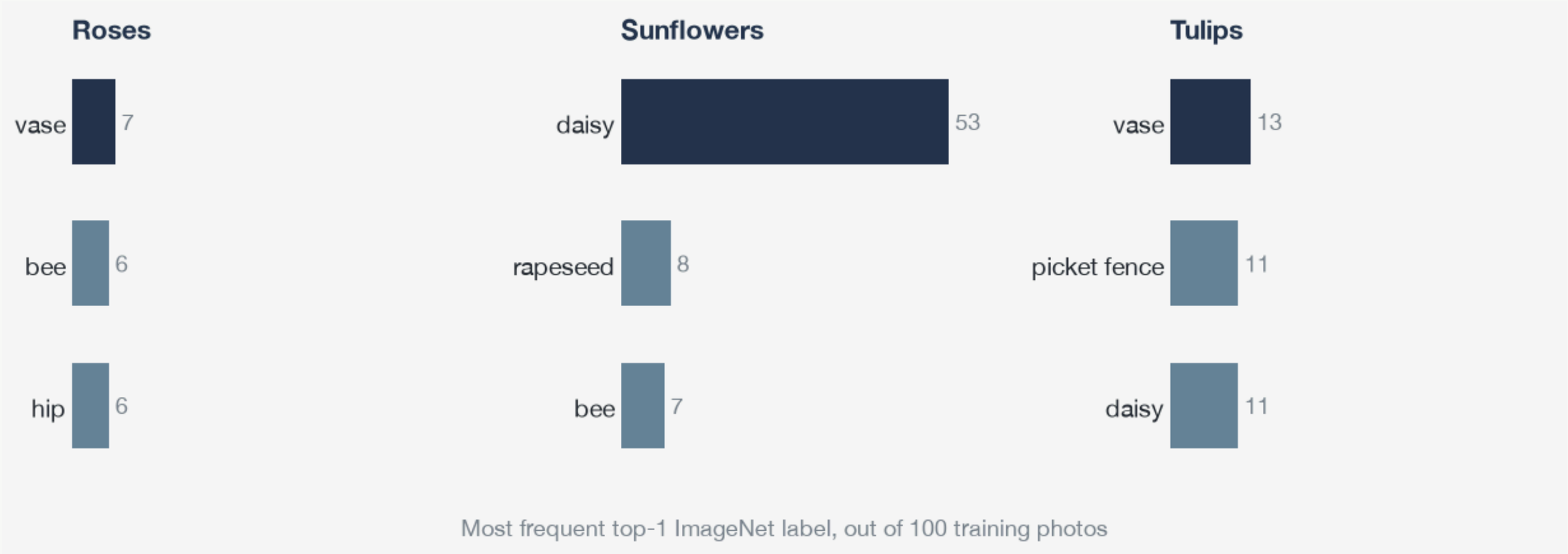
Test photos
used once, at the very end

Source: TensorFlow flower photos (Flickr, CC BY 2.0). Every split has the same number of roses, sunflowers and tulips.

The photos are messy: close-ups, fields and bouquets, 159 to 640 px wide



Off-the-shelf, MobileNetV2 does not know our flowers: its guesses are scattered



ImageNet has no rose, sunflower or tulip class. The only strong pattern is sunflowers read as daisies.

We keep MobileNetV2 as a frozen feature extractor and train only a small classifier

Photo
Resized to 160 × 160 px

→

MobileNetV2, frozen
Pretrained on ImageNet. Turns a photo into 1,280 numbers

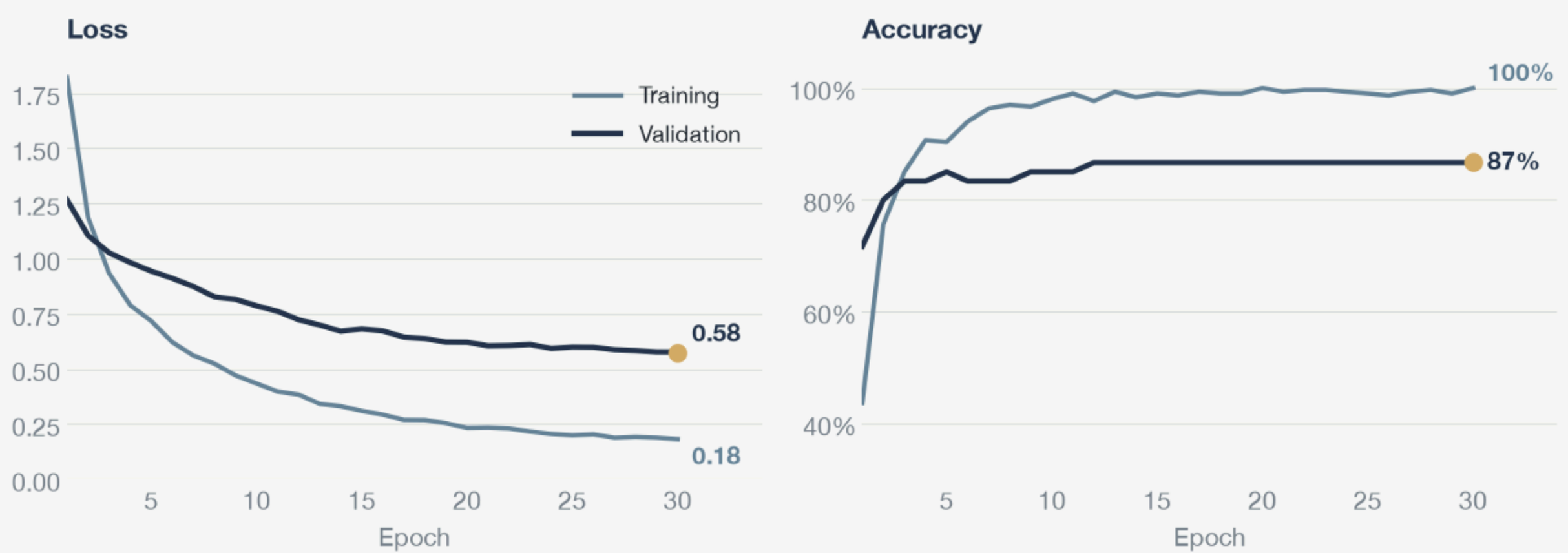
→

Dense layer, trained
Learns three outputs: rose, sunflower, tulip

Model 1: Dense head
Trained on features computed once from the photos

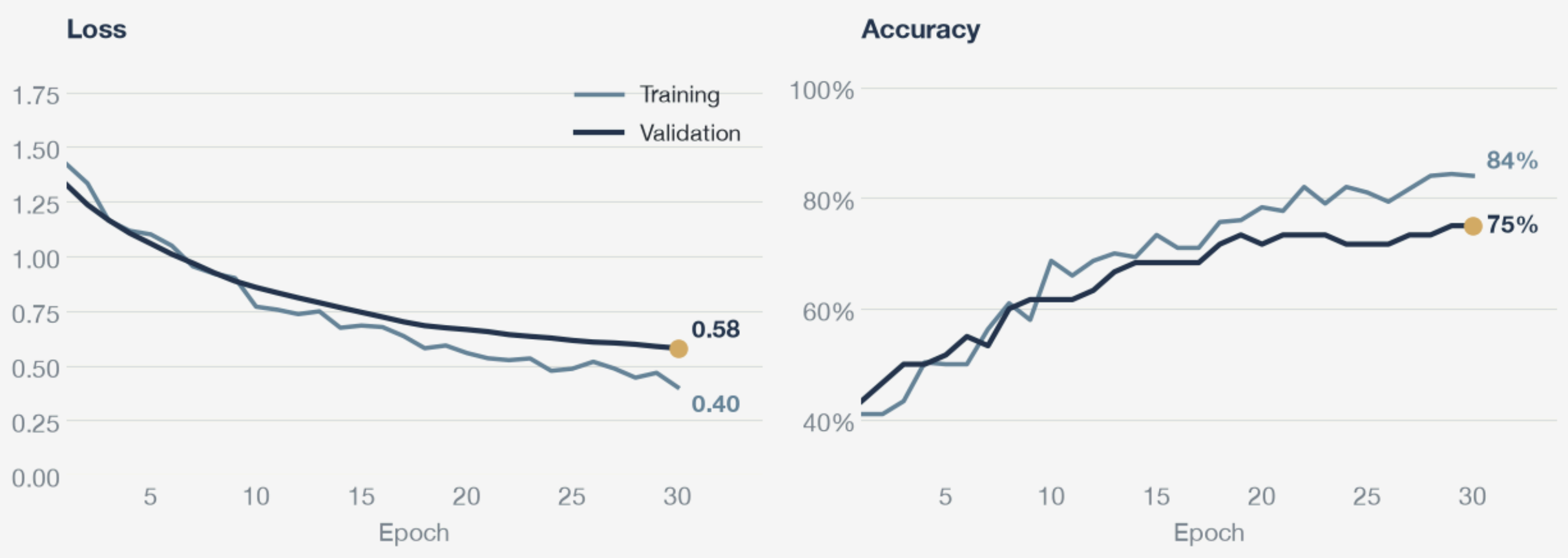
Model 2: Dense head + augmentation
Trained on photos randomly flipped, rotated and zoomed

The dense head memorises the training photos but plateaus at 87% on validation



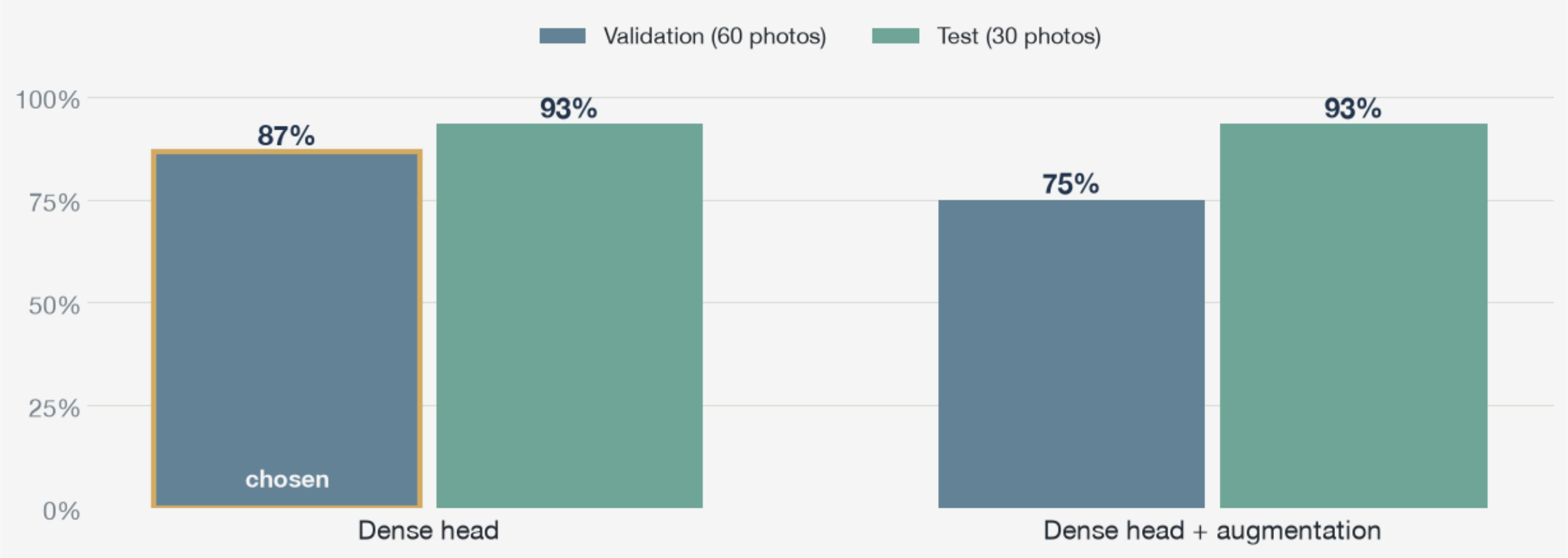
Observed: training accuracy reaches 100%, validation stays near 87%. **Interpretation:** the model overfits a little, as expected with 100 photos per flower.

Augmentation closes the training–validation gap, but validation accuracy drops to 75%



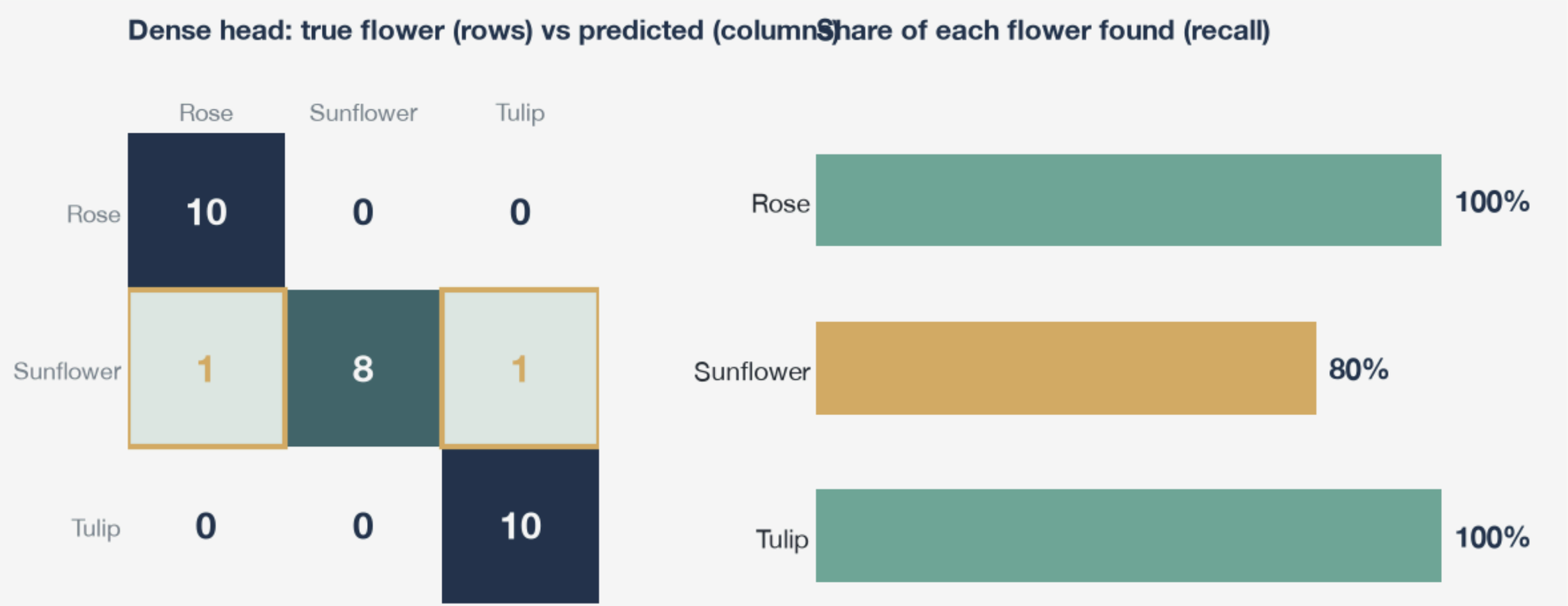
Observed: both curves are still improving after 30 epochs. **Interpretation:** it may need longer training; we did not test that.

The dense head wins on validation; both models classify 28 of 30 test photos correctly



One photo is worth 3.3 points on the test set and 1.7 points on validation, so a gap of two photos (87% vs 93%) is within noise. We chose the model by validation accuracy only.

All of the chosen model's errors are sunflowers: two of ten were mistaken for another flower



The missed sunflowers were not the subject of the photo: a bouquet and a distant garden

True: sunflower Predicted: tulip



True: sunflower Predicted: rose



Observed: in one photo sunflowers sit among other flowers in vases; in the other they are small and far away. **Interpretation:** with only two errors this is a hint, not a proven pattern.

A frozen MobileNetV2 with one dense layer is a strong start, with clear limits

What we found

- **28 of 30** test photos are classified correctly (93%)
- Augmentation did **not** beat the plain dense head on validation
- Both errors were sunflowers in cluttered scenes

What to keep in mind

- 30 test photos is **few**: one photo moves the result by 3.3 points
- Only three flowers are covered; a daisy still gets one of them
- Next: show the covered flowers on the website, let staff review doubtful photos, add more cluttered sunflower examples

Try the model yourself: upload a flower photo and see what it predicts

great-flower-app.streamlit.app

Upload a rose, sunflower or tulip photo and compare the models. Try a cluttered sunflower photo to see where they struggle.